

TP4 filtering

Try to enhance the quality of the signals analysed in TP1 and TP2 :

1. TP2_sig1.txt
2. signal.wav
3. signalAFiltrer.txt from TP1

use pyfdax to generate the coefficients. Compare FIR and IIR approaches.

Exercise 1:

Load TP2_sig1.txt.

1. Make a first filter that extract the sine wave at frequency $f=4/1024$ (normalized)
2. Apply your filter and check the temporal et frequency response
3. Make a second filter that extract the block of sine waves around frequency $f=1001/8192$ (normalized)
4. Apply your filter and check the temporal and frequency response

Exercise 2:

Load the signal signal.wav. (remember to import the library wavfile: *from scipy.io import wavfile*)
`samplerate, y = wavfile.read('signal.wav')`

1. Make a filter that extract the voice only
2. Apply your filter and check the temporal and frequency response
3. save your temporal result as .wav and play it.
`wavfile.write("voiceFiltered.wav", samplerate, y.astype(np.int16))` #save the filtered voice $y(n)$

Exercise 3:

Load the signal signalAFiltrer.txt.

4. Make a filter that extract the low frequency correspondint to a communication signal
5. Apply your filter and check the temporal et frequency response
6. what can we do to improve the result of the filtering ?